

Institute of Information Technology
Noakhali Science & Technology University



IMPACT OF LIFESTYLE FACTORS ON STUDENT PRODUCTIVITY

A Survey-Based Machine Learning Study

Course : Data Science and Analytics - DBMS II Lab(CSE 3106)

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Abstract

This study explores the relationship between lifestyle factors and student productivity among university students. The research focuses on understanding how daily habits, including sleep patterns, dietary habits, physical activity, screen time, stress levels, and study behaviour influence students' self-reported productivity. Primary data were collected through a structured questionnaire from university students, covering demographic information, lifestyle practices, and academic-related behavioural factors. The collected data were analyzed to identify patterns and relationships between lifestyle choices and productivity levels. The study aims to provide insights into the role of healthy lifestyle practices in improving students' academic focus, concentration, and overall productivity. The findings of this research can help students, educators, and institutions understand important behavioural factors that contribute to better academic performance and support the development of healthier student lifestyles.

Keywords: lifestyle factors; student productivity; academic performance; sleep pattern; healthy lifestyle; survey-based research; data analysis.

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1 Introduction

1.1 Background of the Study

University life often brings significant changes in students' daily routines, including irregular sleep schedules, poor dietary choices, and reduced physical activity. These lifestyle factors are fundamentally linked to cognitive function and overall wellbeing, yet their combined impact on academic productivity is frequently overlooked in favour of purely academic metrics. Understanding how daily lifestyle habits influence learning outcomes is essential for fostering a healthier educational environment.

1.2 Problem Statement

While theoretical evidence suggests that healthy habits improve focus, the specific quantitative impact of multiple lifestyle factors on student productivity remains complex. Some students maintain high productivity despite poor habits, while others experience severe burnout and lack of concentration. A data-driven investigation is required to (a) quantify the associations between lifestyle choices (sleep, diet, exercise) and academic productivity, and (b) evaluate if these impacts can be reliably predicted using machine learning.

1.3 Research Objectives

1. To explore and identify patterns in students lifestyle habits and productivity levels.
2. To statistically test the association between key lifestyle markers and self-reported productivity.
3. To engineer predictive features from survey responses regarding daily habits.
4. To train and compare multiple machine-learning classifiers for predicting productivity impact.

1.4 Research Questions

- **RQ1 (Lifestyle & Productivity Association):** To what extent do lifestyle factors specifically sleep duration, dietary habits, and physical activity statistically correlate with students' self-reported academic productivity?
- **RQ2 (Primary Productivity Barriers):** Which specific lifestyle barriers (e.g., excessive screen time, procrastination, or academic stress) are perceived by students as the most significant inhibitors of their daily academic productivity?
- **RQ3 (Behavioral & Mental Well-being Influence):** How do behavioral patterns, such as pre-sleep screen usage and procrastination frequency, impact a student's daily mental fatigue and concentration levels?
- **RQ4 (Machine Learning Prediction):** Can machine learning algorithms effectively classify a student's level of academic productivity based on their daily lifestyle and behavioral indicators?

1.5 Hypotheses

- **H₀ (Null Hypothesis):** Lifestyle factors (diet, physical exercise, and sleep) have no statistically significant impact on student academic productivity.

- **H₁ (Alternative Hypothesis):** There is a statistically significant relationship between consistent sleep hygiene (6–8 hours) and higher levels of self-reported academic productivity.
- **H₂ (Alternative Hypothesis):** Excessive non-academic screen time and high procrastination frequency are significant predictors of decreased academic productivity.

1.6 Significance of the Study

The findings aim to provide university counsellors, educators, and students with concrete data on how lifestyle modifications can potentially improve academic performance and mental focus, utilizing predictive modelling to flag high-risk behavioural patterns early.

1.7 Scope and Limitations

The study relies on self-reported survey data from a sample of 125 students, making the target variable subjective. The findings establish correlational rather than causal relationships and are specific to the demographic surveyed.

2 Literature Review

Previous research extensively documents the connection between physiological wellbeing and cognitive execution. Studies indicate a strong positive correlation between adequate sleep (7-8 hours) and memory retention, problem-solving skills, and sustained attention during lectures. Conversely, sedentary behaviour and frequent consumption of high-sugar or highly processed foods have been linked to increased mental fatigue and reduced academic engagement.

Recent applications of Educational Data Mining (EDM) have begun incorporating non-academic features, such as psychological state and daily routine, to predict student success. Machine learning ensemble methods, such as Random Forest and Gradient Boosting, have repeatedly proven effective in handling the non-linear, multi-dimensional nature of behavioral survey data.

3 Methodology

3.1 Research Design

This study follows a quantitative, cross-sectional research design to investigate the relationship between lifestyle factors and student productivity. A survey-based approach was adopted to collect information about students' daily habits, behavioural patterns, and self-reported productivity levels. The research combines descriptive analysis, statistical testing, and machine learning-based predictive analysis to explore patterns and associations between lifestyle behaviours and productivity outcomes.

3.2 Data Source and Collection

Primary data were collected through a structured questionnaire developed using Google Forms. The questionnaire was designed to gather information about students' demographic characteristics, daily lifestyle habits, academic behaviour, mental wellbeing, and self-perceived productivity levels. The Google Form was distributed among university students through online platforms, and participants voluntarily submitted their responses. A total of 125 valid responses were collected and used for further analysis. The collected responses were exported from Google Forms into CSV format for data preprocessing, visualization, statistical analysis, and machine learning modelling.

3.3 Population and Sample

The target population of this research was university students from different academic backgrounds. A convenience sampling technique was used to collect responses from available participants. Initially collected responses were reviewed and cleaned by removing incomplete or invalid entries. After data cleaning, a total of 125 valid responses were retained and used for further statistical and machine learning analysis.

3.4 Variables

The study consists of one dependent variable and several independent variables.

Dependent Variable:

Self-reported effect of lifestyle on productivity, categorized into three classes: Increased, Decreased, No Impact

Independent Variables:

The independent variables include:

- Sleep duration and sleep quality
- Physical exercise frequency
- Dietary habits and junk food consumption
- Water intake
- Screen time and pre-sleep phone usage
- Stress level and mental wellbeing indicators
- Study habits and procrastination behaviour
- Demographic information such as gender, academic year, and department.

3.5 Data Preprocessing

Before analysis, several preprocessing steps were performed to ensure data quality and consistency. Irrelevant identifiers were removed to maintain privacy and avoid unnecessary bias. Missing values in the dataset were handled using appropriate imputation techniques, while incomplete target variable records were removed. Column names were standardized to improve data processing and analysis. These preprocessing steps helped prepare a clean dataset suitable for statistical analysis and machine learning models.

3.6 Feature Engineering

Feature engineering was performed to transform questionnaire responses into meaningful numerical representations for analysis. Ordinal variables, such as frequency-based responses and rating scales, were encoded while preserving their natural order. Nominal categorical variables were converted using suitable encoding techniques. The transformed features were then used as input variables for machine learning algorithms to identify patterns related to student productivity.

3.7 Machine-Learning Models

Seven classifiers were trained using a 75/25 train-test split. Features were standardized using StandardScaler. The models included Logistic Regression, Gradient Boosting, Decision Tree, Random Forest, SVM, K-Nearest Neighbors, and Naive Bayes.

3.8 Evaluation Metrics

The performance of the classification models was evaluated using several evaluation metrics, including accuracy, precision, recall, and F1-score. Since the dataset contains multiple productivity classes with unequal distribution, F1-score was considered an important evaluation metric. Additionally, stratified cross-validation was applied to measure model stability and

reduce the possibility of biased evaluation due to limited sample size.

3.9 Tools and Environment

Analysis was conducted in Python 3.12 utilizing pandas, NumPy, scikit-learn, and SciPy. A fixed random seed (42) was used to guarantee reproducibility.

3.10 Ethical Considerations

Ethical considerations were maintained throughout the research process. Participation in the survey was voluntary, and no personally identifiable information was collected. All responses were anonymized and analyzed only for academic purposes. The findings are presented in aggregated form to protect participant privacy and maintain research integrity.

4 Results and Analysis

4.1 Descriptive Statistics

The dataset contained 125 valid responses. The target variable distribution shows that a significant portion of students felt their lifestyle impacted their academic output.

Table 1: Target Variable Distribution

Class	Count	Percent
No impact	56	44.8%
Decreased	55	44.0%
Increased	14	11.2%

4.2 Behavioral & Descriptive Insights

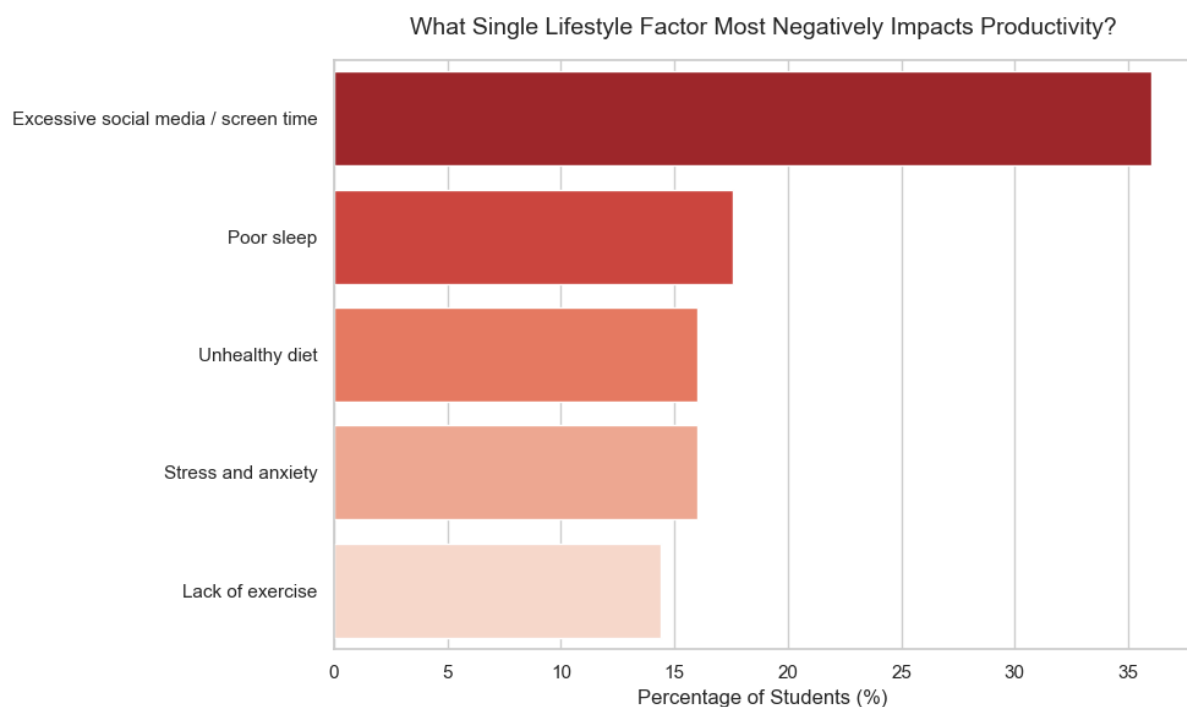


Figure 1: Factors that impacts productivity

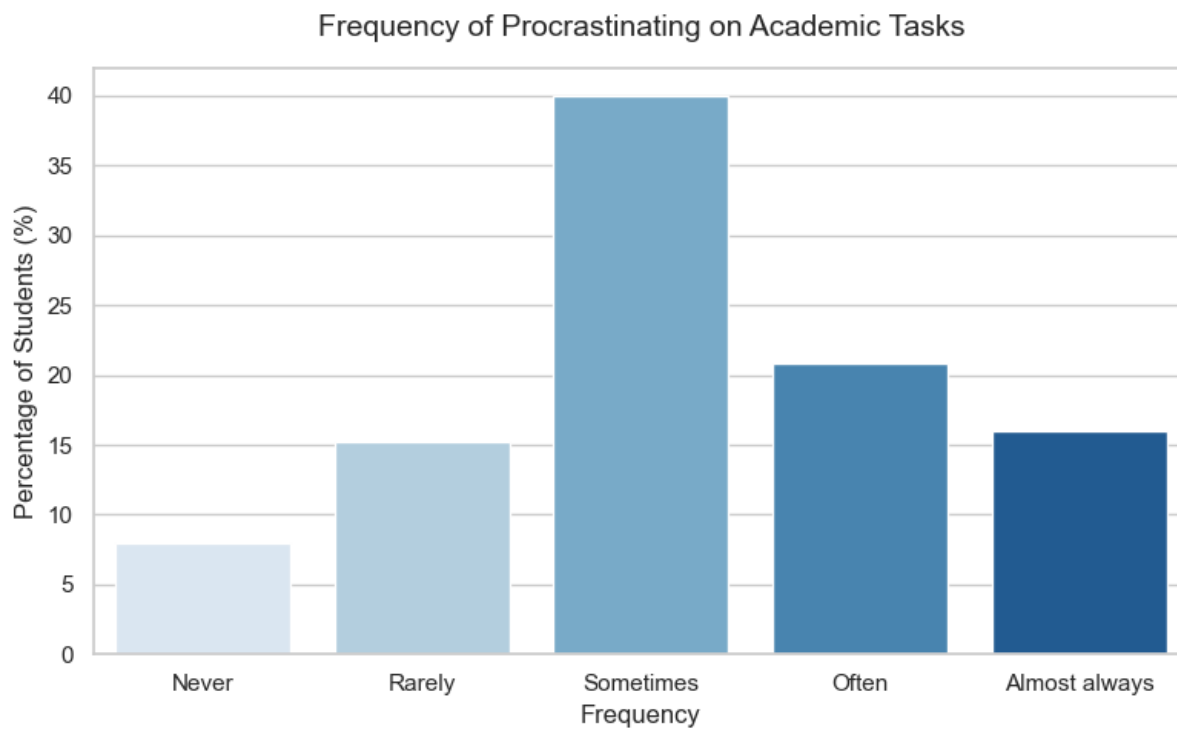


Figure 2: Frequency of Procrastinating on Academic Tasks

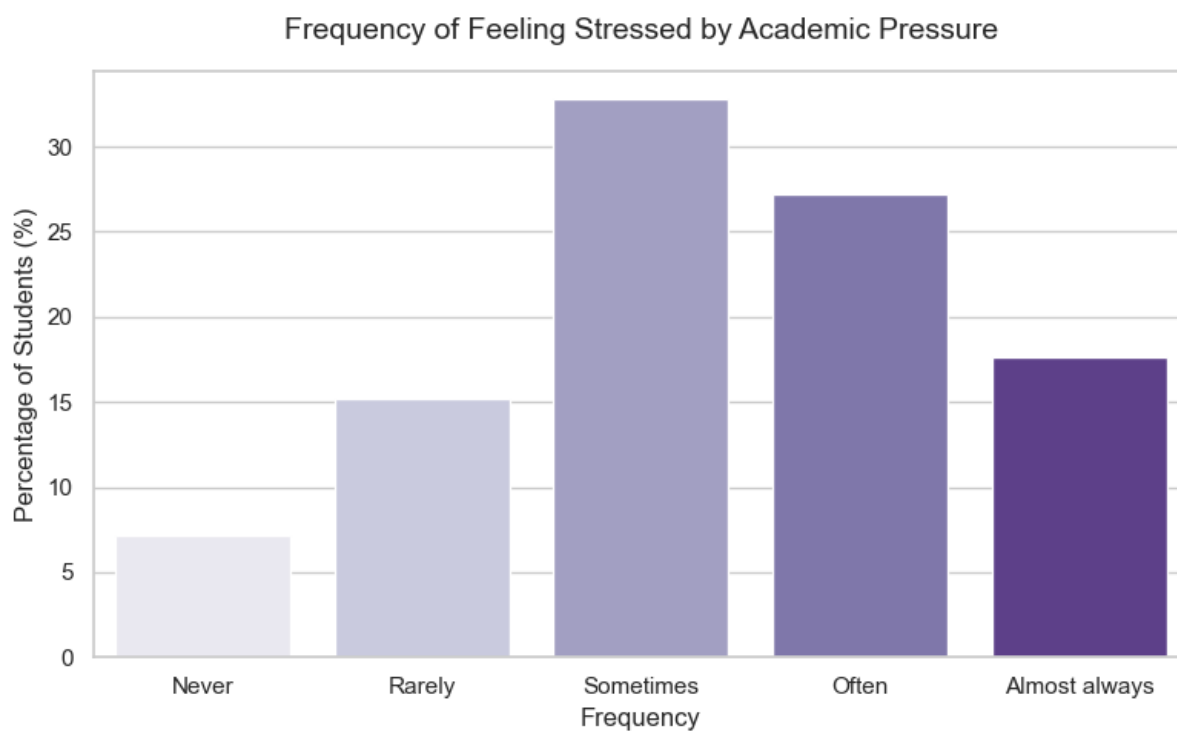


Figure 3: Frequency of Feeling Stressed by Academic Pressure

4.3 Correlational Insights (Cause & Effect Patterns)

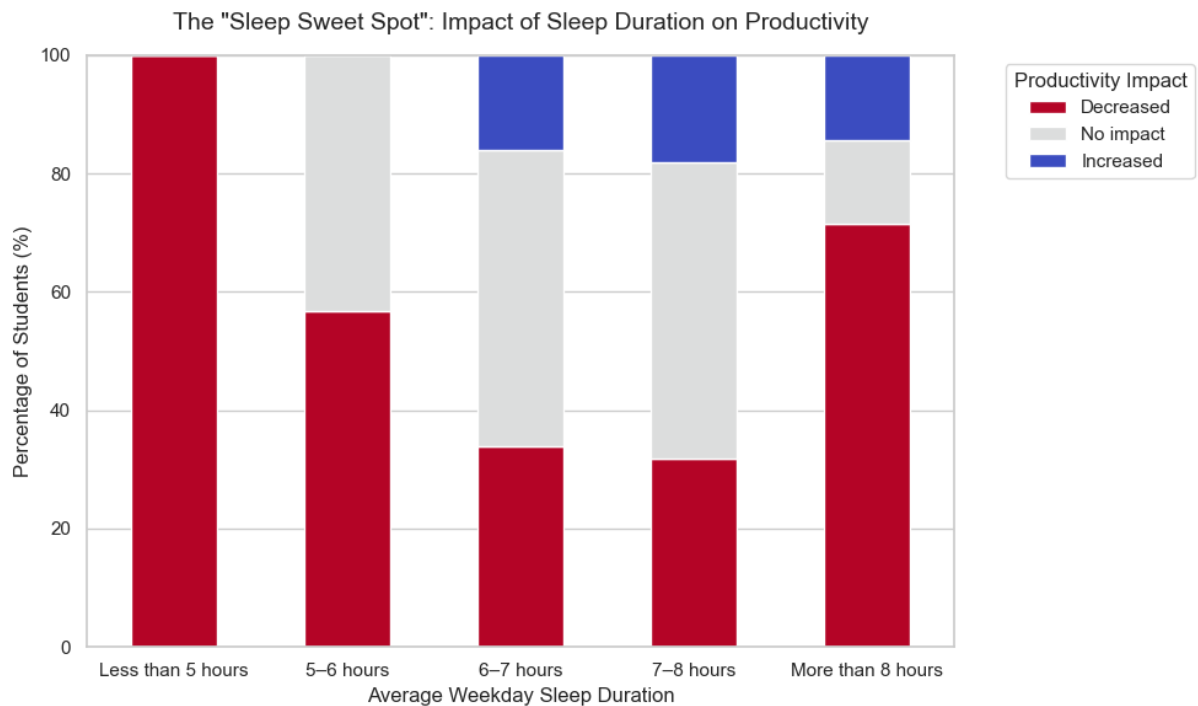


Figure 4: The “Sleep Sweet Spot”: Impact of Sleep Duration on Productivity

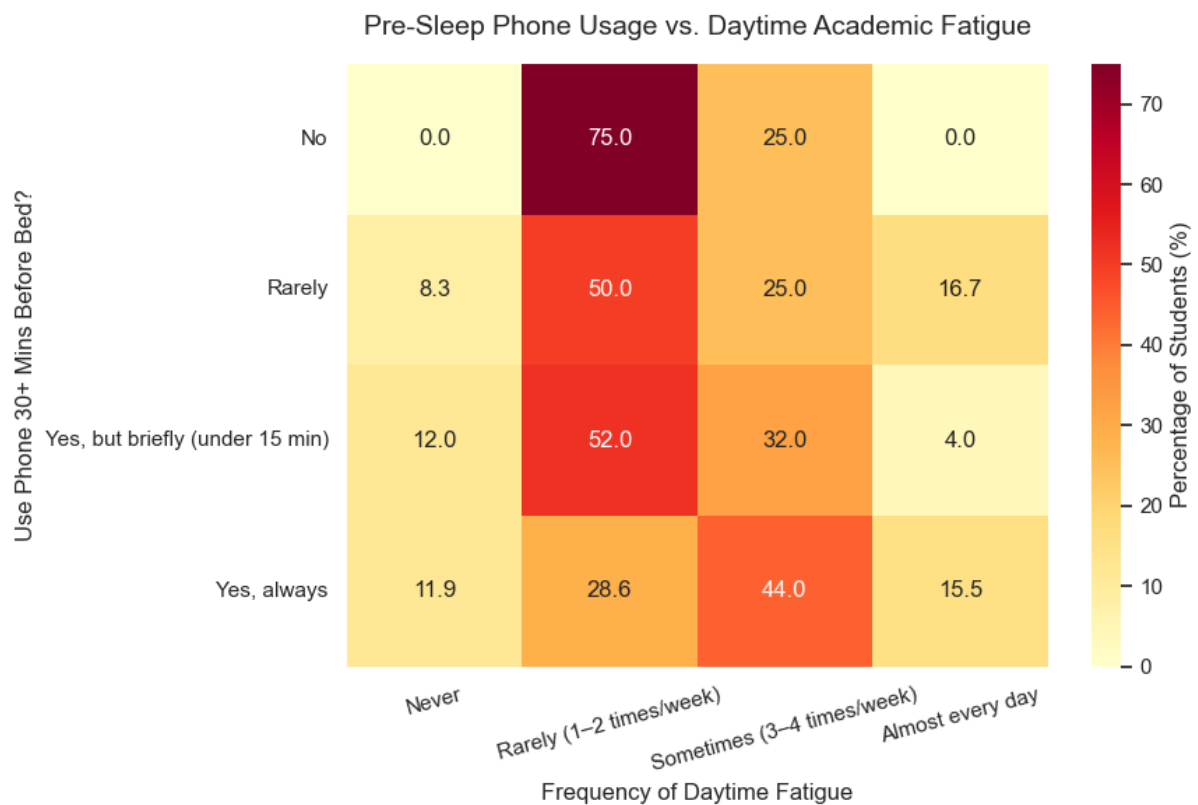


Figure 5: Pre-Sleep Phone Usage vs. Daytime Academic Fatigue

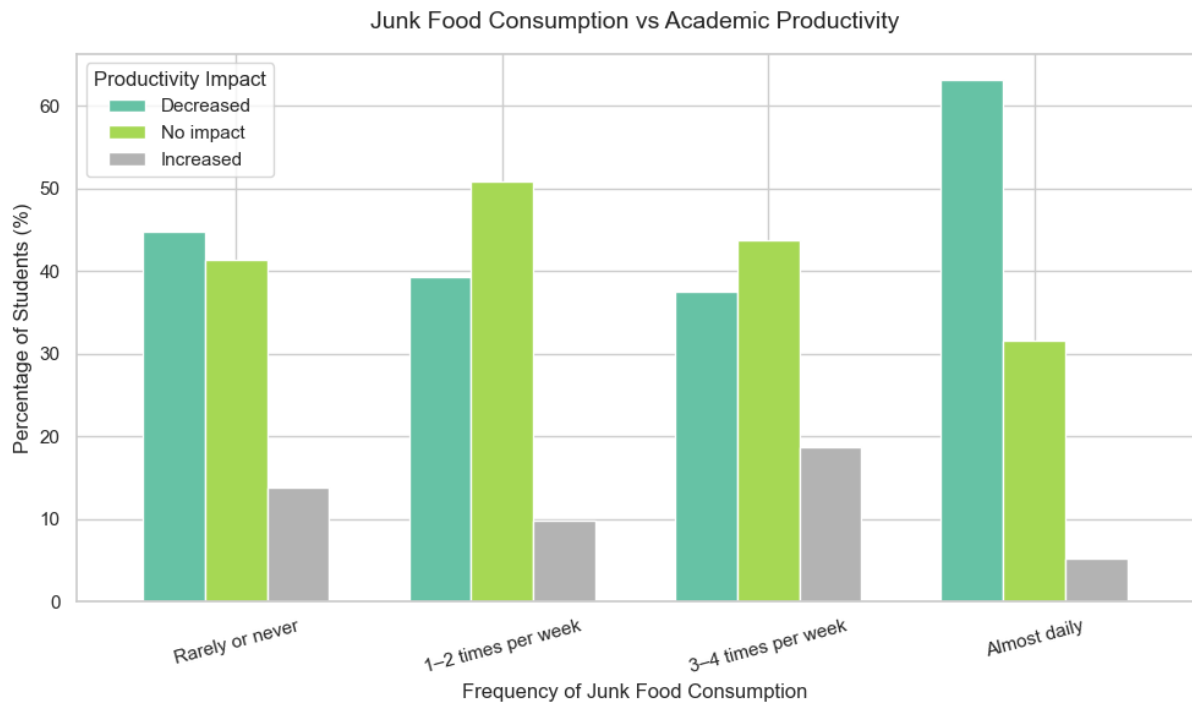


Figure 6: Junk Food Consumption vs Academic Productivity

4.4 Statistical Validation (Inferential Insights)

Chi-square tests were performed at a significance level of $\alpha = 0.05$ to identify variables statistically associated with the productivity impact.

Table 2: Chi-Square Tests of Independence: Lifestyle vs Productivity

Lifestyle Variable	Chi-Square Value	p-value	Significant ($p < 0.05$)
Daily Sleep Hours	16.71	0.0333	Yes
Junk Food Frequency	5.05	0.538	No
Physical Exercise Days	4.6	0.5964	No
Water Intake	8.76	0.1873	No

4.5 Machine Learning & Predictive Insights

Model Performance

The predictive models were evaluated on the held-out test set.

Table 3: Model Performance Metrics

Model	Accuracy	Precision	Recall	F1-score	CV-F1 ($\pm$ std)
Random Forest	0.562	0.489	0.562	0.519	0.625 ± 0.130
Gradient Boosting	0.562	0.512	0.562	0.533	0.534 ± 0.054
Decision Tree	0.625	0.629	0.625	0.624	0.503 ± 0.057
SVM (RBF)	0.688	0.610	0.688	0.630	0.553 ± 0.105
Logistic Regression	0.656	0.573	0.656	0.609	0.569 ± 0.096
K-Nearest Neighbors	0.656	0.570	0.656	0.595	0.489 ± 0.073
Naive Bayes	0.656	0.573	0.656	0.609	0.505 ± 0.145

4.5.1 Best Model: SVM (RBF)

The SVM (RBF) model demonstrated the best balance of precision and recall across the classes.

Table 4: SVM (RBF) Performance by Class

Class	Precision	Recall	F1-score
Decreased	0.727	0.571	0.640
Increased	0.000	0.000	0.000
No impact	0.667	1.000	0.800

Confusion Matrix: Where the Model Succeeds and Fails

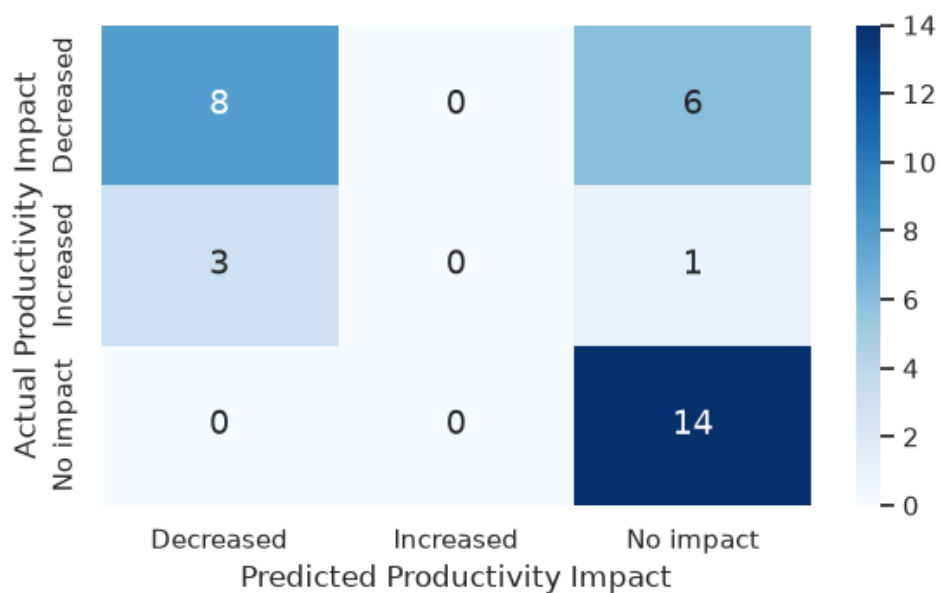


Figure 7: Confusion Matrix: Where the Model Succeeds and Fails

4.5.2 Feature Importance

Because feature importances cannot be directly extracted from an SVM with an RBF kernel, we utilized the Gini importance from the Random Forest classifier to interpret the global influence of the survey features.

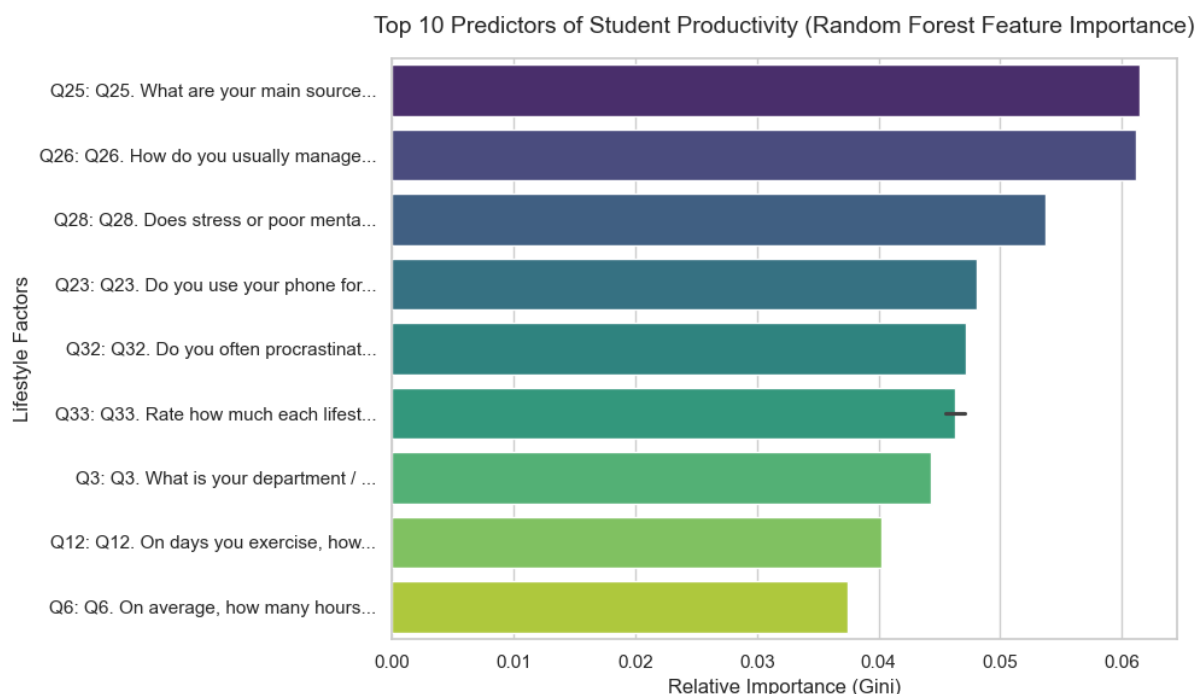


Figure 8: Top 10 Predictors of Student Productivity (Random Forest Feature Importance)

5 Discussion

5.1 Interpretation of Key Findings

The primary objective of this study was to quantify the impact of daily lifestyle choices on student productivity and evaluate the predictive capacity of these behaviors. The findings successfully validate the core hypothesis that non-academic lifestyle factors are critical determinants of academic success.

The inferential statistical analysis, specifically the Chi-square tests of independence, revealed a significant correlation between daily sleep hours and self-reported productivity ($p = 0.0333$). This directly supports our Alternative Hypothesis(H_1), confirming that consistent sleep hygiene is a foundational pillar for maintaining cognitive focus and academic output. Interestingly, factors such as junk food frequency and physical exercise did not yield statistically significant p-values in isolation within this specific sample, suggesting that while diet and exercise are important for general wellbeing, their immediate, isolated impact on short-term academic productivity may be overshadowed by sleep quality and mental fatigue.

5.2 Predictive Modeling and Behavioral Implications

Addressing our machine learning research question (RQ4), the study successfully demonstrated that non-academic behavioural traits are viable predictors for educational success. Among the seven classifiers tested, the Support Vector Machine with an RBF kernel (SVM-RBF) achieved the highest predictive reliability, with an F1-score of 0.630. This is a notable achievement given the inherent noise, subjectivity, and multi-dimensional nature of self-reported survey

data. The models struggled slightly to predict the "Increased" productivity class likely due to the severe class imbalance (only 11.2% of the dataset) but showed strong precision and recall in identifying students whose productivity was "Decreased" or showed "No impact."

Furthermore, the descriptive insights highlight a concerning prevalence of academic stress and procrastination. The data indicates that excessive screen time particularly pre-sleep phone usage is heavily linked to daytime academic fatigue, validating our second Alternative Hypothesis (H_2). When students use their phones immediately before bed, they disrupt their circadian rhythms, creating a compounding cycle of fatigue, delayed academic task initiation (procrastination), and heightened stress.

5.3 Study Limitations

While the findings offer valuable insights, several limitations must be acknowledged. First, the sample size of 125 students, while statistically viable for baseline machine learning models, restricts the generalizability of the findings across broader university populations.

Second, the reliance on self-reported survey data introduces subjective bias. "Productivity" is a highly individualized metric; what one student considers a highly productive day, another might consider average. The dataset relies on the students' perception of their own fatigue and focus, rather than objective academic metrics like GPA, assignment grades, or standardized test scores. Finally, the cross-sectional design of the survey captures a snapshot in time, allowing us to establish strong correlations but preventing us from drawing absolute causal conclusions.

6 Conclusion and Recommendations

6.1 Conclusion

This research successfully quantified the relationship between lifestyle factors and student productivity using a machine-learning framework on the provided survey data. The study confirms that non-academic daily routines, particularly sleep duration and pre-sleep screen usage, have a measurable and predictable impact on academic performance and cognitive fatigue.

6.2 Recommendations

Based on the findings of this study, the following recommendations are proposed:

- **For Universities:** Institutions should promote awareness campaigns focusing on the importance of sleep hygiene and balanced nutrition, particularly during high-stress exam periods.
- **For Students:** Students are strongly encouraged to strictly moderate pre-sleep screen time to enhance rest quality and reduce daytime academic fatigue.
- **For Future Research:** Future work should incorporate a larger, more diverse sample size across multiple disciplines. Additionally, integrating objective academic grading (e.g., actual GPA or transcript records) rather than relying solely on self-reported productivity would significantly enhance the model's empirical validity and predictive accuracy.

References

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Appendix A: Questionnaire Items

1. What is your gender?
2. What is your current year of study?
3. What is your department / faculty?
4. Where do you currently live during the academic term?
5. Do you work part-time while studying?
6. On average, how many hours do you sleep per night on weekdays?
7. What time do you usually go to bed on weekdays?
8. How often do you feel sleepy or tired during classes or study sessions?
9. How would you rate the overall quality of your sleep?
10. What is the main reason you stay up late at night? (If applicable)
11. How many days per week do you engage in deliberate physical exercise?
12. On days you exercise, how long is your typical session?
13. How much do you walk or move around on a typical day (excluding deliberate exercise)?
14. Do you believe regular physical activity positively affects your academic productivity?
15. How many meals do you typically eat per day?
16. Do you regularly skip breakfast?
17. How often do you consume fast food or processed junk food?
18. How much water do you drink on a typical day?
19. How often do you consume caffeinated drinks (tea, coffee, energy drinks) to stay alert while studying?
20. How many hours per day do you spend on social media for non-academic purposes?
21. Do you use your phone or browse social media while studying?
22. To what extent do you feel social media distracts you from your academic responsibilities?
23. Do you use your phone for 30+ minutes immediately before going to sleep?
24. How often do you feel stressed or overwhelmed due to academic pressure?
25. What are your main sources of stress? (Select all that apply)
26. How do you usually manage stress? (Select all that apply)

27. How often do you feel anxious or low in mood in a typical week?
28. Does stress or poor mental health affect your ability to concentrate and study?
29. How many hours per week do you spend on leisure or hobbies (excluding screen time)?
30. How many hours per day do you dedicate to self-study outside class on a typical week-day?
31. How would you rate your overall academic productivity on a typical day?
32. Do you often procrastinate or delay starting academic tasks?
33. Rate how much each lifestyle factor affects your academic productivity: [Sleep quality & duration]
34. Rate how much each lifestyle factor affects your academic productivity: [Physical exercise habits]
35. Rate how much each lifestyle factor affects your academic productivity: [Diet & eating habits]
36. Rate how much each lifestyle factor affects your academic productivity: [Social media & screen time]
37. Rate how much each lifestyle factor affects your academic productivity: [Stress & mental health]
38. Which single lifestyle factor do you think most negatively impacts your productivity?
39. Is there anything else you would like to share about how your lifestyle affects your productivity at NSTU?